

Exercise 1:

The result of a survey conducted by a certain oil company on the evolution of the price x_i of an oil barrel and the number of oil barrels sold y_i , are given in the following table:

Unit price of the oil barrel: x_i in dollars	97	98	99	100	101	102
Number of oil barrel sold: y_i in millions	20	19.5	18.8	17.4	16.9	16.7

In this exercise, given your answers rounded to the nearest 10^{-2} .

- Construct, in an orthogonal system, the scatter plot of the points (x_i, y_i) .
- Calculate the coordinates of the center of gravity G , and plot this point in the preceding system.
- Calculate the coefficient of correlation and interpret the value obtained.
- Write an equation of regression line $(D_{y/x})$ of y in terms of x , and plot this line in the preceding system.
- Suppose that the evolution of the price follows the given pattern and that the price of an oil barrel reaches 105\$.
 - Show that the estimation of the number of oil barrels sold, at this price, is 14.62 million.
 - The production cost of one oil barrel is 25\$. Estimate the total profit in this case.

Exercise 2:

The table below shows the production of apples (in ton), in a small village, in the years 2006. Till 2012

Year	2006	2007	2008	2009	2010	2011	2012
Rank of the year x_i	1	2	3	4	5	6	7
Production of apples (in ton) y_i	3.5	3.8	4.6	6.5	6.9	7.8	9

- Calculate the means $\bar{x}$ and $\bar{y}$ of the variables x & y .
- Represent graphically the scatter plot of the points $(x_i ; y_i)$ as well as the center of gravity G in a rectangular system.
- Calculate the correlation coefficient r and give an interpretation of the value obtained.
- Determine an equation of regression line $(D_{y/x})$, and draw this line in the preceding system.
- Suppose that the above pattern remains valid till 2020.
 - Estimate the production of apples in year 2018.
 - In which year the production will exceed 12.5 tons.

Exercise 3:

The table below shows the math grade of 9 students and their rank in class.

Grade x_i	15	10	8	6	7	4	6	9	11
Rank y_i	11	12	9	5	6	7	6	8	10

1. Represent graphically, the scatter plot of the points $(x_i; y_i)$, as well as the center of gravity G in an orthogonal system.
2. Calculate $\text{cov}(x; y)$ and the correlation coefficient e of this distribution. Interpret.
3. Determine an equation of the regression line $(D_{y/x})$, y in terms of x , and plot it in the preceding system.
4. Estimate the rank of the student who got 13 on the math test.

Exercise 4:

The table below represents the number of internet clients from the year 2000 to 2009.

Year	2000	2002	2004	2006	2009
Rank of the year x_i	1	3	5	7	10
Number of the clients y_i (in thousands)	0.5	3	6	8	12

1. Calculate $\bar{x}$ and $\bar{y}$ the respective averages of x and y .
2. Represent graphically the scatter plot of the points $(x_i; y_i)$, in an orthogonal system.
3. Determine the equation of the regression line $(D_{y/x})$, of y in terms of x and construct this line in the same system.
4. Estimate the number of internet clients for the year 2012.
5. Starting from which year will the number of internet clients exceed 20 000?
6. Calculate the percentage of increase of the number in clients from the year 2004 to 2006
7. In reality the number of internet clients in the year 2012 is 16000.

Calculate the percentage of error between reality and the estimated value.

Exercise 5:

The following table shows, for each year, the sales of a company (in millions LL).

year	2008	2009	2010	2011	2012	2013	2014
Rank x_i of the year	1	2	3	4	5	6	7
Sales y_i (in millions LL)	12	19	29	37	45	53	62

- 1) a- Draw, in a rectangular system, the scatter plot of the points associated to the distribution $(x_i; y_i)$.
b- Calculate the coordinates of the center of gravity G and plot this point in the preceding system.
c- Determine an equation of the regression line $(D_{y/x})$ of y in terms of x , and draw this line in the same system.
- 2) Calculate the correlation coefficient r of this distribution, and interpret the value thus obtained.
- 3) Assume that the above pattern remains valid for the next ten years.
 - a- Estimate the sales of the company in the year 2018.
 - b- Estimate the percentage of increase in the sales from the year 2014 till 2018.